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Beyond pixels: Zero-shot classification of chest X-ray reports via prompt engineering and LLMs

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ABSTRACT

Accurately extracting structured clinical labels from unstructured chest radiology reports, particularly the MIMIC-CXR dataset, is critical for robust Clinical Decision Support Systems. However, conventional supervised deep learning faces major challenges: reliance on expert-annotated datasets, which are costly and time-consuming to curate, and the computational overhead of training or fine-tuning complex models. To circumvent these limitations, this study proposes a Prompt based Zero Shot Learning Framework that leverages the semantic reasoning capabilities of pre trained LLM to classify radiology reports without requiring gradient-based training or labeled data. We systematically investigate the impact of various text preprocessing pipelines and prompt engineering strategies, including task specific prompts, Persona based instructions, Chain of Thought reasoning, and Ensemble Prompting, alongside mechanisms for managing label uncertainty. Experimental results demonstrate that the optimal configuration, utilizing the IMPRESSION section of reports with specific uncertainty handling, achieves a weighted average F1 score of 0.85 and an Area Under the Curve of 0.92. These results significantly outperform traditional rule based labelers and unsupervised baselines in the domain of clinical text classification. This study substantiates the potential of NLP and prompt engineering to automate medical report structuring, offering a scalable and label free alternative to traditional supervised learning for healthcare applications.

1. Introduction

The effective utilization of clinical reports and patient notes recorded by medical staff is pivotal for enhancing both the speed and accuracy of disease diagnosis and patient management in healthcare facilities [1]. Automating the analysis of these documents can significantly reduce the administrative burden on human resources and optimize clinical workflows [2]. Consequently, there is a growing demand for systems capable of processing unstructured patient narratives to classify new cases into established disease categories.

The development of clinical Natural Language Processing (NLP) models, particularly those based on deep learning, typically necessitates large-scale labeled clinical datasets [3,4]. However, the data labeling process represents a significant bottleneck. Such datasets are rarely publicly available due to privacy concerns, and their annotation requires the expertise of radiologists or clinicians, incurring substantial costs and time. Furthermore, given the complexity of medical data and stringent regulatory requirements, dataset curation often spans several months [5].

To surmount these challenges, Zero-Shot Learning (ZSL) techniques and pre-trained Deep Learning models have emerged as powerful alternatives, enabling the classification of samples from novel classes without prior specific training [6]. In the medical domain, where accuracy is directly linked to patient health outcomes, model reliability is paramount. Prompt-based learning has surfaced as a technique to maximize this reliability. By designing and fine-tuning customized prompts, we can modify the input format to align with the pre-trained knowledge of Large Language Models (LLMs), thereby eliciting highly accurate outputs [7].

LLMs have become transformative technologies in NLP, demonstrating immense potential at the intersection of Artificial Intelligence (AI) and Biomedical and Health Informatics (BHI) [8]. Their capacity to process complex linguistic structures allows them to interpret diverse medical data, including electronic health records and clinical notes, with remarkable proficiency [9,10]. However, the probabilistic nature of LLMs means their outputs may contain degrees of uncertainty. Managing this uncertainty is a critical aspect of deployment.

This study specifically focuses on the classification of chest

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radiography reports. Chest X ray imaging offers distinct advantages, including widespread availability, cost effectiveness, and rapid acquisition [5]. It remains a primary diagnostic tool for pulmonary abnormalities such as pneumonia, tuberculosis, and COVID 19 [4,11]. Foundational research has demonstrated the efficacy of Convolutional Neural Networks for binary classification tasks like COVID 19 detection [12,13] and provided comprehensive surveys of deep learning applications in this domain [14]. Building on this, recent advancements have led to highly accurate diagnostic models using architectures like MaxViT [15], Swin Transformers [16], and Optimal Cross Stage Partial Bi LSTM networks [17]. However, despite the vast volume of imaging data generated annually, the scarcity of radiologists available to interpret these images and the computational cost of training these complex vision models create significant diagnostic bottlenecks. Consequently, there is an urgent need for AI driven Clinical Decision Support Systems (CDSS) to assist medical professionals [18].

Our research targets the classification of textual reports associated with these images, specifically using the MIMIC-CXR Database v2.1.0 [19]. Because radiology reports are unstructured, they cannot be directly processed by standard algorithms; entities and concepts must first be extracted using NLP techniques [20]. While prior studies have made strides in identifying chest pathologies, they often face significant limitations: reliance on large-scale supervised training, high computational costs, and difficulties in generalizing to new data formats [21]. Furthermore, multi-modal models like CLIP often require paired image-text data that may not exist for specialized domains.

In this proposed method, we leverage the advanced instruction-following capabilities of LLMs within a Zero-Shot framework. This approach eliminates the need for model training from scratch, thereby removing the dependency on labeled datasets. By systematically evaluating various prompt engineering strategies, we identify the optimal interaction protocols to maximize diagnostic accuracy. The results demonstrate that our text-based approach not only bypasses the costs of image analysis but also achieves performance superior to existing baselines, offering a robust solution for automated clinical report structuring.

Beyond standard pathology classification, the flexibility of this Zero Shot, prompt-based methodology unlocks numerous clinical and operational applications. In real world settings, this framework can facilitate Dynamic Triage, prioritizing reports not just by fixed labels but by urgency (e.g., prompting the model to identify "reports requiring immediate intervention"). Furthermore, it serves as a robust tool for Quality Assurance, enabling automated verification of whether a radiologist's "Impression" fully captures the key "Findings." In the research domain, this approach revolutionizes Cohort Discovery, allowing investigators to query massive, unstructured archives using natural language prompts (e.g., "extract all reports with incidental pulmonary nodules") to rapidly build study cohorts for retrospective analysis without the need for fragile, custom built NLP pipelines.

The specific contributions of this study are as follows:

- Systematic Ablation Study: A comprehensive analysis of various configurations, including text preprocessing pipelines, prompt engineering strategies, and uncertainty management techniques for MIMIC CXR classification.
- Modality Shift: A novel focus on the classification of CXR samples using only textual reports, eliminating the computational overhead of image analysis.
- Label Efficiency: The elimination of training costs and the dependency on labeled datasets by employing a zero-shot classification framework.
- Clinical Viability: The demonstration of high accuracy results (F1 score 0.85), indicating the capability of the system to serve as a reliable assistant in Clinical Decision Support Systems.

- Generalizability: A dataset agnostic framework that leverages the semantic reasoning of LLMs rather than statistical fitting, ensuring applicability to other clinical domains.
- Benchmarking LLMs: Providing comparative evidence that pre trained LLMs, when properly prompted, outperform traditional supervised text classifiers in pathology detection.

2. Related works

Given the necessity and significance of identifying chest-related diseases and clinical conditions for timely treatment [22], numerous studies have been conducted in this field. The objective of this section is to review approaches and methods utilizing machine learning and deep learning techniques that can serve as a foundation for future studies and research in chest disease detection. Generally, previous studies can be categorized into different groups based on the type of work done, as shown in Table 1.

2.1. Multi-modal and image-based approaches

Several studies have focused on leveraging both visual and textual

Table 1
Categorization of previous studies based on the type of work conducted.

Category	Strengths	Weaknesses	References
Text and Image Models	Understanding semantic similarities between images and related texts, no need for labeled data	Requires specific image-text data for a particular domain	[23–26]
Zero-shot	Superior performance in identifying rare diseases in MIMIC-CXR reports, better outcomes in some cases compared to supervised methods	—	[23, 25–28]
Few-shot	Better results in some cases compared to transfer learning methods	—	[24]
Fine-tuning	Improved results on text reports, enhancement with three data types (image, text, label), better performance in analyzing specific disease severity	Lack of diverse data for precise fine-tuning in some specific tasks	[26,29,30]
Prompt Learning	Addressing dependency on handcrafted prompts, increased generalizability	—	[26,28]
Prompt Engineering	Optimized communication with LLMs and generating appropriate responses	—	[28]
LLMs	No need for human-labeled data, faster inference process, increased generalization capability, integration of medical knowledge into prompts	—	[28]
Dataset Creation	Includes 3D CT chest images, reduced errors in disease diagnosis and interpretation	Limited data diversity, need to employ other 3D imaging methods like MRI and PET	[27]
Multi-modal	Modeling various data types in a shared feature space, addressing data scarcity issues, improved diagnosis accuracy, better performance compared to single-modal models	Lack of diverse data, GPU memory limitations	[26,29]
MIMIC-CXR Dataset Analysis	Data preparation for training a LLM	—	[31]

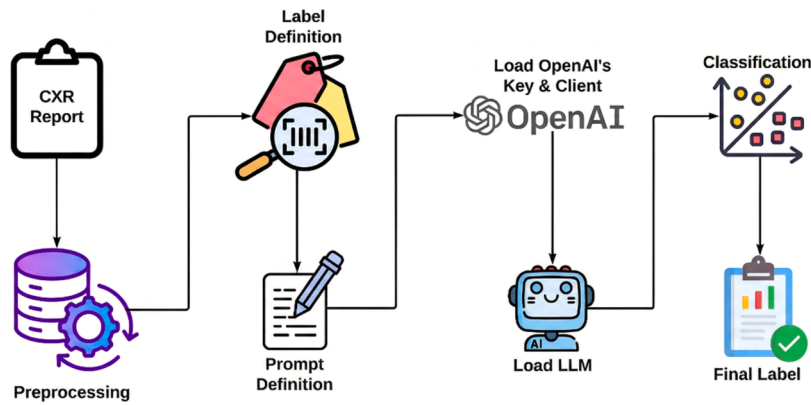


Fig. 1. Workflow of the proposed method for classifying CXR textual reports.

data. For instance, FlexR [24] introduced a framework for structured reporting using CLIP. However, its reliance on a few-shot paradigm necessitates access to high-quality labeled samples for fine-tuning, limiting its utility in strictly zero-shot scenarios. Similarly, CheXzero [32] and Mishra et al. [25] demonstrated the power of contrastive learning for pathology detection. While effective for common diseases, these hybrid architectures often struggle with domain shift and require massive paired image-text datasets for pre-training.

To address false negatives in contrastive learning, Jang et al. [23] proposed a fine-tuning strategy using random text sampling and relaxation loss. However, this approach still operates within a multi-modal alignment framework, requiring computational overhead for image encoding. More complex frameworks like UMCL [26] integrate three data types (image, text, label) into the pre-training objective. While innovative, this increases computational complexity.

Furthermore, Hamamci et al. [27] introduced a foundation model utilizing chest CT volumes and radiology reports for zero-shot detection. While this advances multi-modal capabilities into 3D imaging, it necessitates the curation of massive, complex volumetric datasets. Our work challenges this high-resource necessity. Our work challenges this multi-modal necessity, investigating whether the textual report alone processed by advanced LLMs can provide a more efficient and equally accurate diagnostic signal without the need for image data or complex pre-training pipelines.

2.2. Text-based and NLP approaches

Parallel to image-based work, progress has been made in analyzing radiology reports using NLP. RC-TPL [28] employed prompt learning to align report-level and concept-level features. However, this approach relies on a complex pipeline involving explicit entity extraction (e.g., via RadGraph), which can be fragile if the extraction step fails. Similarly, the MoRE framework [29] improves accuracy by combining X-rays, ECGs, and text, but the authors acknowledge that GPU memory constraints limited their ability to utilize scalable Large Language Models (LLMs).

In the domain of pure text classification, Uslu et al. [30] established a strong baseline by fine-tuning BERT-based models on the MIMIC-CXR dataset. While effective, their results suggested a performance plateau, indicating that discriminative encoders may have reached their limit without extensive supervised training. Additionally, Uslu et al. [31] provided a comprehensive semantic and structural analysis of MIMIC-CXR reports, utilizing methods like NER and n-gram analysis. While this offers valuable descriptive insights for data preparation, it does not in itself provide a classification solution. Our research bridges these gaps by replacing the supervised fine-tuning and manual feature extraction paradigms with a generative Zero-Shot framework,

leveraging the emergent reasoning capabilities of modern LLMs to achieve superior performance without labeled training data.

2.3. Cross-domain strategies and research gap

It is worth noting that the challenge of extracting structured intelligence from unstructured reports and handling data scarcity is not unique to medicine but extends to high-stakes domains like cybersecurity. Recent advancements such as CSKG4APT [33] and ThreatInsight [34] have demonstrated the efficacy of Knowledge Graphs in structuring dispersed threat data, while AEKG4APT [35] utilizes LLMs to automate this knowledge extraction, validating the capability of LLMs to parse complex, domain-specific narratives. Furthermore, architectures like the Dynamic Prototype Network (DPNSA) [36] have addressed data scarcity via few-shot learning.

Our work parallels these methodological advancements but introduces a distinct paradigm shift to address the limitations of current medical AI:

1. **Zero-Shot vs. Fine-Tuning:** Unlike BERT-based approaches [30] that require extensive labeled datasets for fine-tuning, our framework operates in a strict Zero-Shot setting, eliminating the dependency on costly expert annotation.
2. **Text-Centric Efficiency:** In contrast to multi-modal frameworks like CheXzero [32] or BioViL [37] that require computationally intensive image processing, we demonstrate that textual reports alone, when processed by generative LLMs, yield superior classification performance ($F1 = 0.85$).
3. **Systematic Optimization:** Rather than proposing a single model, we provide a comprehensive ablation study of 27 configuration strategies, revealing that data selection (e.g., using the "IMPRESSION" section) is more critical for performance than prompt complexity.

3. Proposed method

This study leverages the capabilities of pre-trained LLMs, specifically the GPT family, due to their high accuracy and compatibility with the prompt-based zero-shot learning approach. One of the advantages of these models is their ability to handle new tasks by optimizing and customizing prompts, allowing users to structure and detail their requests in a way that yields appropriate outputs. This enables the model to classify samples from unseen classes and adapt to new scenarios. The objective of this research is to classify textual reports derived from chest radiology images, using data collected from the MIMIC-CXR Database.

In the proposed method, the following steps are performed sequentially: Preprocessing, Label Definition, Prompt Definition, LLM Loading, and Sample Classification. Finally, the results are evaluated. The

complete end-to-end workflow of this Zero-Shot classification framework is formally outlined in Algorithm 1. The findings indicate that the proposed approach achieves higher accuracy compared to previous methods, eliminating the need for model training and saving both time and resources. The workflow diagram of the proposed method and the preprocessing steps for samples are illustrated in Figs. 1 and 2, respectively. Further details of the proposed method are provided in the subsequent sections.

Algorithm 1. Zero-Shot Classification of CXR Reports

	Require: R: The set of all CXR report texts. L: The list of target clinical labels. C: The experimental configuration
	Output: Y_pred: A list of predicted label sets, one for each report.
1	Initialize LLM_client with model="gpt-4o-mini", temperature=0.
2	Initialize Y_pred = [].
3	
4	for each report_text in R do
5	// Step 1: Text Selection and Preprocessing
6	selected_text = SelectSection(report_text, C.text_section)
7	processed_text = Preprocess(selected_text)
8	
9	// Step 2: Prompt Construction
10	final_prompt = ConstructPrompt(processed_text, L, C.prompt_template)
11	
12	// Step 3: Zero-Shot Inference
13	response = LLM_client.query(final_prompt)
14	predicted_labels = ParseLabels(response)
15	Y_pred.append(predicted_labels)
16	end for
17	
18	return Y_pred

3.1. Preprocessing

Text data preprocessing reduces the volume of input samples, enabling the scalability of computationally expensive models and facilitating the processing of larger datasets [38]. In this study, various common preprocessing techniques in NLP were reviewed to clean and refine the data. Approaches that demonstrated positive effects on improving model performance and results were identified and adopted as the final preprocessing steps in the proposed method. The specific sequence of these operations is critical for maintaining semantic integrity, as illustrated in Fig. 2. For instance, Abbreviation Expansion is performed first to resolve acronyms like "PA" (Posteroanterior) before Lowercase Conversion, ensuring that capitalization cues are not lost. Following this, Noise and Punctuation Removal strip non-informative characters to focus the model's attention on clinical entities.

Abbreviation Expansion: This technique is an important step in preprocessing text reports. It involves identifying clinical abbreviations and replacing them with their full forms, helping to achieve a more accurate interpretation of the text. Many medical reports contain numerous abbreviations, and if these are not properly expanded, the model may fail to understand their true meaning. This can negatively

affect tasks such as classification, clinical information extraction, and question-answering models [39]. In the proposed method, the Spacy library and ScispaCy models, a customized version of Spacy for analyzing scientific texts in fields like medicine and biology, are used [40]. This Python library is designed for NLP and offers capabilities such as POS tagging, NER, and dependency parsing. The procedure is as follows: a dictionary for abbreviation expansion is created using a list of x-ray imaging abbreviations collected at the University of Saskatchewan, Canada. The text of the CXR report is then tokenized using the "en_core_sci_sm" model, which is designed for scientific and specialized texts. Each token is checked, and if it matches an abbreviation in the

dictionary, its expanded form is substituted in the input text.

Noise Removal: In this method, all unnecessary non-ASCII characters that do not contribute to improving model performance are removed using a regular expression pattern. This reduces complexity, simplifies the text, and increases processing speed by reducing data volume [41].

Punctuation Removal: Using the *string.punctuation()* function in Python, punctuation marks such as "!", "?", ".", and control characters like Tab or newline characters are removed from the input text. The presence of these punctuation marks and control characters may not add useful information and can introduce noise. Some punctuation marks can also lead to incorrect tokenization. By removing them, the data is cleaned, allowing the model to focus on key words and main concepts [42]. This helps the model focus more on relevant data and avoid processing extraneous and irrelevant information.

Lowercase Conversion: In clinical texts, a word may appear in various forms, such as "Pneumonia," "pneumonia," and "PNEUMONIA." The lowercase conversion technique is one of the most common methods for normalizing text. By eliminating case sensitivity, it can improve the performance of text classification tasks [43]. Since most clinical abbreviations appear in uppercase in the text, the process of converting tokens to lowercase is done after expanding clinical abbreviations to prevent errors and the loss of important information in this context [42].

To increase the comprehensiveness of the evaluation, several NLP

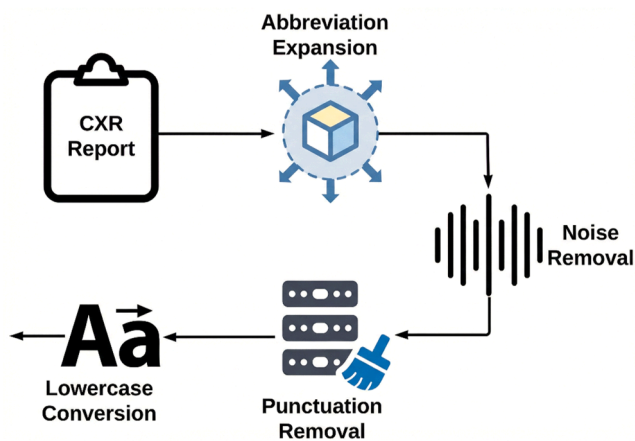


Fig. 2. Data preprocessing steps in the proposed method.

Table 2
Various strategies in the preprocessing stage.

Step	Preprocessing Method
1	Noise Removal, Punctuation Removal, Lowercase Conversion
2	Step 1 + Sentence-level Tokenization
3	Step 1 + Stop Word Removal
4	Step 1 + Number Removal
5	Step 1 + Sentence-level Tokenization
6	Step 1 + NER
7	Abbreviation Expansion + Step 1

preprocessing methods and various combinations of them were used, and the results were analyzed. However, some of these methods did not have a significant impact on the model's output. The details of these methods are explained below, and additional information is provided in Table 2.

Sentence Tokenization: This technique is a process in NLP that divides text into sentences. It is important for working with long texts. After this step, sentences are considered as separate units that can be processed independently. Clinical reports are usually lengthy and complex, containing multiple sentences with medical details, key findings, and related interpretations. Dividing the text into sentences helps to analyze each sentence individually. Each sentence in a medical report may refer to a specific part of the body, such as "increased opacity in the left lower lobe" or "no pleural effusion observed". Sentence-level tokenization helps to better analyze each sentence.

Remove Stop Words: Stop words typically do not differentiate between different text documents [44], so their removal is a common preprocessing step used to reduce data size and focus on key words. Medical reports often contain sections with minimal informational content, such as "The findings are consistent with..." In this example, removing stop words like "The", "are", and "with" makes the text more concise and focused. This is especially important in large models and with large datasets.

Remove Numbers: Removing numbers that do not carry significant informational value is a common text normalization technique used to simplify the content [45]. For example, "Admission Date: [2021–12-09]". In this case, the numerical values are not useful for predicting or describing the patient's clinical condition. On the other hand, some numbers in medical reports are vital, and removing them could result in the loss of important information. For instance, "Pleural effusion measures 2.5 cm". Here, the value "2.5 cm" contains crucial clinical information, or numbers related to the size of masses or laboratory values. Given this importance, a customized approach for removing numbers can be used. Some strategies include:

- Only remove irrelevant numbers, i.e., analyze semantically beforehand to check whether the number is linked to vital information. For example, dimensions, clinical values, and percentages should be retained, while hospital codes, serial numbers, and dates can be removed.
- Use pre-trained NLP models to distinguish numbers related to clinical contexts.
- Instead of completely removing numbers, they can be replaced with a placeholder to retain their significance, such as "<number>".

NER: Named Entity Recognition plays a crucial role in the clinical and biomedical field, with the goal of classifying entities in text [46]. The use of NER in radiology reports can assist in identifying the following types of information [47]:

- **Identification of Medical and Diagnostic Terms:** This includes diseases and conditions such as "pleural effusion," "pneumonia," "cardiomegaly," or symptoms and findings like "opacity," "nodule," "consolidation."
- **Identification of Anatomical Locations:** These include terms such as "right lung," "mediastinum," or "left diaphragm," which can be important for diagnosing the location of lesions or injuries.
- **Identification of Modifiers:** Phrases such as "mild," "severe," or "progressive," which describe the severity or characteristics of a disease.

Thus, Named Entity Recognition can help transform radiology report texts into a structured format. Details of all preprocessing approaches in the proposed method are reported in Table 2.

3.2. Label definition

The MIMIC CXR dataset comprises 14 distinct labels: 12 corresponding to specific chest pathologies, one for "Support Devices" (referring to implanted medical equipment), and a "No Finding" label indicating the absence of any other identified pathology. In this study, we focus on a 13 class subset of the dataset, specifically by excluding samples where "No Finding" is the sole positive label (i.e., instances where "No Finding" equals 1). During the Label Definition stage, this refined set of 13 target labels is formally provided to the model. Consequently, in the prompt presented to the LLM, it is instructed to review the sample and output all relevant labels from this defined set that match the patient's condition

3.3. Prompt definition

Prompts are instrumental in the proposed method, serving as the interface between CXR textual reports and the pre trained Large Language Model. In NLP tasks, prompt definition involves crafting a specific text fragment, or prompt template, to elicit a desired output from the model [48]. The efficacy of zero shot inference hinges on selecting appropriate text prompts that clearly define classes, provide explicit instructions, and integrate the sample text for analysis [49]. Large Language Models are generally pre trained to generate responses based on given instructions and context, leveraging extensive datasets and state of the art deep learning techniques [50]. Through meticulously designed prompts, our approach effectively extracts key information from radiology reports, defining the task with clarity and precision. For example, instructing the model to classify medical reports into specific categories such as "Pleural Effusion" or "Pneumothorax" clearly articulates the model's objective and structures its output. LLMs, with their vast knowledge base from diverse scientific and medical texts, utilize prompts to align their general understanding with specific task requirements.

In Zero Shot Learning, the model is expected to execute the requested task without any retraining on new domain specific data. Prompts are

crafted to equip the model with sufficient information regarding the data structure and the target classes. Well-structured and precise prompts are essential for minimizing model errors and producing reliable outputs. For instance, specifying the output format as "label1, label2, label3, ..." helps the model generate predictable and relevant responses, avoiding extraneous information.

Prompt engineering refers to the systematic process of designing and creating prompts to extract desired answers from LLMs. These prompts are categorized based on characteristics such as structure, functionality, and complexity. Each prompt is formulated as a natural language question to obtain accurate and specific responses from a pre trained language model [51]. Prompt engineering enables the selection of the most appropriate prompt template for a specific task [52]. In this study, examples of practical prompts will be reviewed and explained in detail:

3.3.1. Task-specific prompt

Task-specific prompts are designed to perform a particular task or function and typically instruct the model to carry out a specific action or analysis. These prompts might include tasks such as diagnosing a disease from a medical report, translating a sentence, simulating a particular style of writing, or classifying samples. In other words, these prompts are tailored to achieve a well-defined objective that the model must accomplish. Their structure often includes the following elements [20]:

- **Instructions:** Define the task, explain the nature of the data, and outline the expected outcomes from the model.
- **Input Arguments:** Provide the essential information required to perform the task.
- **Constraints and Output Format Examples:** Guide the model in generating appropriate and structured outputs.

The majority of the prompts examined in this research are task-oriented. Additionally, some techniques and other approaches are incorporated into these prompts, the details of which will be discussed further in the study.

3.3.2. Using persona in prompts

Incorporating a persona into prompt design involves assigning a role to the AI model, allowing it to act as a character or system aligned with a specific task or domain [51]. This technique, combined with providing detailed instructions, enables the model to deliver more relevant, accurate, and focused responses by considering the context and specific needs of the user [53]. This approach is widely used in prompt design and interaction with LLMs, guiding the model's responses to align with a predefined set of attributes, behaviors, or traits. It is systematically applied in prompt engineering processes to create structured interactions [54]. For example, if the model assumes the role of a "radiologist," its responses will lean toward specialized analysis and interpretations in the medical imaging field.

3.3.3. Few-shot prompts

Few-shot prompts enable LLMs to perform various tasks with limited training data. Instead of fine-tuning the model with a large dataset, this approach presents a few examples of task-specific input-output pairs (e.g., 1–100 examples per class) within the prompt. This helps the model understand the task's context and response format. Few-shot learning is particularly valuable in the field of clinical NLP due to the scarcity of labeled training data. In medicine, especially concerning rare diseases, collecting large volumes of annotated textual data by experts is costly and time-consuming. Therefore, algorithms capable of learning from a small number of examples can revolutionize this domain [55]. Other approaches based on few-shot learning include One-Shot Prompting, where only one example is provided. This method is mostly suitable for simpler tasks. Additionally, Zero-Shot Prompting does not include any examples; the model must generate responses solely based on the task description. The advantages of few-shot prompting include reducing the

need for large, labeled datasets, enabling the model to learn diverse tasks with just a few examples, eliminating the need for fine-tuning and computationally intensive processes, and allowing for faster experimentation.

3.3.4. Chain of thought (CoT)

CoT prompts assist models in implementing step-by-step reasoning processes. In this approach, the model is instructed to articulate its reasoning incrementally rather than providing an immediate answer. This method allows the model to tackle complex problems by breaking them down into intermediate steps and solving them progressively [56]. Unlike traditional methods that focus solely on delivering answers, CoT emphasizes the hierarchy of the reasoning process that leads to the solution [57]. This helps the model utilize available information more clearly and logically, reducing the likelihood of errors in the reasoning process. It is particularly effective for tasks requiring intricate reasoning [58]. To further enhance the CoT process, multiple examples are sometimes used to demonstrate how step-by-step reasoning is performed, enabling the model to "think" more effectively. These prompts are widely applied in solving mathematical and logical problems that require systematic, step-by-step reasoning.

3.3.5. "Take a deep breath" technique

This technique can be utilized in prompt design to create a space for more thoughtful analysis by the model. In situations where problems are complex or require detailed evaluation, this prompt helps the model avoid making hasty and superficial decisions. Instead, it encourages the model to analyze the content carefully, leading to more meaningful responses at each step [59,60]. In this approach, the prompt is usually designed to invite the model to think systematically and step-by-step. This method is particularly useful for complex tasks in NLP, such as classification or identifying specific features in data.

3.3.6. Scene defining

This technique involves crafting prompts that define the context or environment in which the task is performed [61]. It helps the model understand the scenario and the nature of the task more accurately. For example, in the context of classifying CXR reports, such a prompt might define the scene details as follows: "In a radiology department, classify the following findings from the CXR using the provided labels. If necessary, select multiple relevant labels. Provide your response strictly in the format label1, label2, label3, without any additional explanations or spaces before and after the response." Here, phrases like "In a radiology department" and "findings from the CXR" serve as contextual details and scene descriptions for the model. This approach helps the model narrow its focus and correctly interpret aspects related to the clinical report.

3.3.7. Ensemble prompting

In this approach, a set of prompts is used to improve the accuracy and reliability of the model. Instead of relying on a single prompt, multiple distinct prompts are designed, each guiding the model toward the correct answer from a different perspective or method. The final answer is then derived by combining the outputs of these prompts. Different prompts can highlight various aspects of the problem, resulting in a more comprehensive response. This approach leverages the strengths of diverse prompts while mitigating their weaknesses [51]. Ensemble prompting can also enhance the stability of the model's responses; if outputs across different prompts are consistent, the likelihood of their correctness increases. The process involves presenting each prompt individually to the model and collecting the resulting outputs. These outputs are then combined using techniques such as majority voting, weighted averaging, or statistical analysis. In the proposed method, a combination of three prompts "instruction-based using persona", "few-shot" and "CoT" was employed. The final response was determined through a majority voting approach, where the labels with the highest frequency were considered the final output. Details of the reviewed

Table 3
Overview of the proposed method's prompts with examples.

Index	Prompt text	Type
1	As a radiologist, classify the following radiology report with the labels in the list of classes You may select multiple labels that are most relevant Return the output strictly as label1, label2, label3, with no additional explanations or spaces before or after the response Labels: [cxr_labels] radiology report: [cxr_text]	Using persona
2	As a radiologist, classify the given CXR report provided below, with the labels in the list of classes. You may select multiple labels that are most relevant. Return the output strictly as label1, label2, label3, ... with no additional explanations or spaces before or after the response. Example 1: [example1] Labels: Atelectasis, Cardiomegaly, Consolidation, Enlarged Cardiomeastinum, Lung Opacity, Pleural Effusion, Pleural Other, Pneumonia, Pneumothorax, Support Devices Example 2: [example2] Labels: Atelectasis, Consolidation, Edema, Fracture, Lung Lesion, Pleural Effusion, Pneumonia, Support Devices Example 3: [example3] Labels: Cardiomegaly, Enlarged Cardiomeastinum, Lung Opacity, Pleural Other, Pneumothorax List of classes: [cxr_labels] CXR report: [cxr_text]	Few-shot
3	As a skilled radiologist, carefully analyze the following CXR report. Identify and select the most relevant labels from the provided list. Labels: [cxr_labels] CXR Report: [cxr_text] Output: Please provide a comma-separated list of labels without any additional text or spacing. For example: label1, label2, label3	Using persona
4	Task: Classify the Chest X-ray report. Input: [cxr_text] Labels: [cxr_labels] Output: A comma-separated list of relevant labels.	Simple Task-specific
5	As a radiologist, carefully classify the following Chest X-Ray report using the provided list of labels. Think step by step: 1. First, review the report content in detail. 2. Identify specific findings and match them with the labels in the list. 3. Select the most relevant labels that match the report findings. Multiple labels can be chosen if necessary. Return your answer strictly as label1, label2, label3, ... with no additional explanations, spaces, or punctuation. List of Labels: [cxr_labels]. CXR Report: [cxr_text].	CoT
6	Take a deep breath and approach the following chest X-ray report analysis calmly. As a radiologist, read through the report with attention to detail. Identify the key findings and choose the labels from the provided list that best describe the report. - Focus on each detail in the report, matching it carefully to the list of labels. - Select all relevant labels, as some cases may require multiple choices. Respond with the labels only, in this exact format: label1, label2, label3, ... with no extra explanations or spaces. Labels Available: [cxr_labels]. Report: [cxr_text].	Take a deep breath
7	As a skilled radiologist, carefully classify the following Chest X-Ray report using the provided list of labels. Think step by step: First, review the list of provided	CoT

Table 3 (continued)

Index	Prompt text	Type
8	labels and ensure you understand the definition of each. Next, compare the findings in the report to the descriptions of the labels. If there is a clear match, select the corresponding label. If there is uncertainty, consider consulting additional resources or seeking a second opinion. List of Labels: [cxr_labels]. CXR Report: [cxr_text]. Return your answer strictly as label1, label2, label3, ... with no additional explanations, spaces, or punctuation. As a skilled radiologist, systematically analyze the following CXR report. Think step by step. Begin by breaking down the report into sections based on anatomical regions (e.g., lungs, heart, mediastinum). For each section, carefully examine the findings described, paying attention to key terms such as opacity, nodule, effusion, and consolidation. Once you have a thorough understanding of the findings in each section, integrate the information to arrive at a comprehensive interpretation and select the most relevant labels from the provided list. List of Labels: [cxr_labels]. CXR Report: [cxr_text]. Return your answer strictly as label1, label2, label3, ... with no additional explanations, spaces, or punctuation.	CoT
9	In a radiology department, classify the following chest X-ray findings using the labels from the provided list. Select multiple relevant labels if necessary. Return your answer strictly as label1, label2, label3, with no additional explanations or spaces before or after the response. Labels: [cxr_labels]Radiology report: [cxr_text]	Scene defining
10	Ensemble of prompts 1,2,5	Ensemble
11	prompt = " Classify the following radiology report with the labels in the list of classes. You may select multiple labels that are most relevant. Return the output strictly as label1, label2, label3, ... with no additional explanations or spaces before or after the response. Labels: [cxr_labels]. Radiology report: [cxr_text]." Response = client.chat.completions.create(model=" gpt-4o-mini", messages = [["role": "system", "content": "You are a radiologist tasked with classifying radiology reports based on provided labels."], ["role": "user", "content": prompt]], max_tokens=100, temperature=0, n = 1,)	Using system role

prompts, along with examples, are provided in [Table 3](#).

3.4. Implementation details

In this study, based on the documentation provided by OpenAI, the GPT 4o mini model was selected for categorizing textual reports of CXR images due to its optimal balance of accuracy, high speed, and cost efficiency compared to other models on the platform. Access to OpenAI's service was established through a secure API key, enabling the transmission of requests and retrieval of model responses. The selected model was integrated into the proposed framework using OpenAI's Python API functions, facilitating dialogic interaction with the model.

The proposed method utilizes a Zero Shot approach for classifying samples, leveraging engineered prompts and LLMs. This strategy eliminates the need for training or fine tuning, obtaining desired outputs simply through well designed prompts. The primary parameters and configurations are outlined below:

- Model: The specific model identifier used was gpt 4o mini.

- **Messages:** The request structure consists of a list of messages defining the interaction context. Each message includes two main components: "role" and "content". The role designates the sender and can be "system", "user", or "assistant". These roles guide the model's behavior:
- o **System Role:** Provides general instructions and defines the model's "persona." For instance: ["role": "system", "content": "You are a helpful assistant specialized in medical imaging."]. This instructs the model to adopt the behavior of a medical imaging specialist, enforcing a specific framework for its responses.
- o **User Role:** Contains the specific input from the user. In this study, the "user" role encapsulates the engineered input prompt and the specific radiology report content to be classified.
- o **Assistant Role:** Represents the format of a desired response. This is particularly useful in few shot scenarios, where sample responses are provided to guide the model toward the correct output structure.
- **max_tokens:** Determines the maximum length of the generated response. To ensure focused classification outputs and optimize costs, this value was set to 100 tokens.
- **temperature:** Controls the randomness of the output. For this study, to ensure deterministic and reproducible results, the temperature was set to 0. While higher values (>0) introduce variability useful for creative tasks, clinical classification requires the high probability precision achieved at zero temperature.
- **n:** Specifies the number of response candidates to generate. This was set to 1 to obtain a single, deterministic prediction. While generating multiple responses ($n > 1$) can be beneficial for exploring diversity, it increases computational cost and requires additional selection logic.

After initializing the LLM with these configurations, the pre-processed samples and selected prompts are transmitted to the model. The resulting response, containing one or more predicted labels, is parsed and stored as a structured list for each sample. Finally, the model's performance is evaluated by comparing these predicted lists against the ground truth.

3.4.1. Experimental environment and specifications

All experiments were conducted using the Google Colab cloud platform. The client-side scripts were executed in a Python 3.10 environment. Given the API based nature of our approach (accessing the gpt 4o mini model), heavy computational tasks (specifically the Large Language Model inference) were offloaded to OpenAI's servers. Consequently, no high-end local GPUs were required for the inference phase.

- **Software and Libraries:** The implementation relied on the openai Python library (version 1.14.0) for API interaction, pandas for data manipulation, and scikit learn for calculating evaluation metrics (Accuracy, F1 score, AUC).
- **Hardware (Client Side):** The Google Colab environment provided an Intel(R) Xeon(R) CPU @ 2.20 GHz and approximately 12 GB of RAM, which was sufficient for text preprocessing and API request management.

Table 4
Statistical analysis of dataset samples based on token count.

	Full Dataset			Random 10,000 samples		
	Findings & Impression	Findings	Impression	Findings & Impression	Findings	Impression
Number of Samples	59250	81855	113507	10000	10000	10000
Total Initial Tokens	6508424	6332813	5557376	1046520	732879	427593
Avg. Initial Tokens	109.84	77.36	48.96	104.65	73.28	42.75
Total Preprocessed Tokens	5908098	5862729	5076853	937577	670274	384553
Avg. Preprocessed Tokens	99.71	71.62	44.72	93.75	67.02	38.45

Table 5
Percentage frequency of labels in the 10,000 random sample subset.

Label	FINDINGS & IMPRESSION	FINDINGS	IMPRESSION
Atelectasis	36.49 %	39.38 %	37.02 %
Cardiomegaly	18.76 %	28.25 %	28.21 %
Consolidation	7.08 %	5.66 %	8.59 %
Edema	24.60 %	22.56 %	28.10 %
Enlarged Cardiomediastinum	1.84 %	3.19 %	4.01 %
Fracture	3.45 %	3.57 %	3.68 %
Lung Lesion	5.59 %	4.87 %	4.76 %
Lung Opacity	38.46 %	38.91 %	33.35 %
Pleural Effusion	34.35 %	37.75 %	38.56 %
Pleural Other	0.88 %	0.83 %	1.07 %
Pneumonia	28.72 %	19.08 %	24.29 %
Pneumothorax	4.41 %	6.14 %	6.58 %
Support Devices	7.56 %	22.21 %	24.99 %

- **Network:** Data transmission was conducted over standard secure HTTPS protocols to communicate with OpenAI API endpoints. No specialized local network hardware was utilized, highlighting the accessibility and low resource requirements of our proposed Zero Shot framework.

4. Experimental settings

4.1. Dataset

4.1.1. Dataset description and experimental design

This study utilizes the MIMIC-CXR Database v2.1.0 [19], a large-scale collection of chest radiographs and their corresponding free-text radiology reports. The dataset contains 227,835 studies. For our classification task, we focused on 13 pathological labels, excluding the "No Finding" class. The labels are encoded as: 1 (Positive), 0 (Negative), and -1 (Uncertain).

Our experimental design involved two key decisions. First, to manage uncertain labels, we evaluated two common binarization strategies: U-Ones (mapping $-1-1$) and U-Zeros (mapping $-1-0$). Second, since the "FINDINGS" section contains granular observations while the "IMPRESSION" provides a synthesized summary [30], we evaluated the performance on these sections both individually and in combination to identify the most signal-rich text.

4.1.2. Token and label distribution

Due to the significant computational cost and API rate limits associated with LLM inference, experiments were conducted on a randomly selected subset of 10,000 samples.

The total token count varies significantly by section, impacting processing efficiency. The complete dataset contains over 23 million tokens, whereas using only the "IMPRESSION" section reduces this to approximately 5.5 million. A detailed breakdown of token counts is presented in Table 4.

The label distribution within this subset was analyzed to ensure it mirrors the natural class imbalance of the full dataset. As shown in Table 5, the subset captures the full spectrum of label frequencies: from

high-prevalence classes (e.g., "Atelectasis") to rare pathologies (e.g., "Pleural Other"). This distributional alignment confirms that no sampling bias was introduced.

4.1.3. Statistical justification of evaluation subset

To ensure the reliability of our results, we conducted a statistical power analysis. For the full MIMIC-CXR population ($N \approx 377,000$), a random sample size of $n = 10,000$ provides a Margin of Error of less than 1 % at a 99 % Confidence Level. This confirms that the performance metrics derived from this subset are generalizable to the entire dataset with high confidence. Furthermore, this evaluation scale significantly exceeds standard benchmarks in the field; for comparison, the official CheXpert test set contains only 500 images [28].

4.2. Evaluation metrics

To rigorously assess the model's performance in the sensitive domain of clinical classification, we employ a comprehensive suite of standard metrics: Accuracy, Precision, Recall, F1-score, and Area Under the Curve (AUC). Given the significant class imbalance inherent in the MIMIC-CXR dataset (as detailed in Section 4.1.1), standard accuracy can be misleading. Therefore, this study prioritizes the **Macro-Averaging** technique. This method calculates metrics for each label independently and then averages them, ensuring that rare pathologies (minority classes) contribute equally to the final score as frequent ones, preventing bias toward majority classes.

4.2.1. Accuracy

This metric represents the percentage of all samples that the model has correctly classified, whether positive or negative. The calculation method is shown in equation (1).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

4.2.2. Precision

The ratio of correctly predicted positive observations to the total predicted positives. This metric is crucial in clinical settings where False Positives (misdiagnosis) can lead to unnecessary stress and costs.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

4.2.3. Recall (Sensitivity)

The ratio of correctly predicted positive observations to the all observations in the actual class. In medical screening, high Recall is often prioritized to minimize False Negatives (missed diagnoses).

$$\text{Recall} = \text{TPR} = \frac{TP}{TP + FN} \quad (3)$$

4.2.4. F1-score

The harmonic mean of Precision and Recall. This metric is particularly useful for imbalanced datasets as it seeks a balance between false positives and false negatives.

$$F1 = 2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

4.2.5. AUC (Area Under the Curve)

AUC (Area Under the Curve): The area under the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. AUC measures the model's ability to distinguish between classes, ranging from 0.5 (random guess) to 1.0 (perfect discrimination). It is a robust

metric for evaluating classifier performance across different decision thresholds [62].

4.3. Evaluation protocol

This study employs a Zero-Shot Learning framework, which fundamentally differs from traditional supervised learning approaches. Consequently, there is no conventional training or fine-tuning process on the MIMIC-CXR dataset. Instead, our "validation procedure" is a direct Zero-Shot Evaluation Protocol, performed as follows:

1. **Evaluation Dataset Selection:** As detailed in Section 4.1, a random subset of 10,000 samples was selected from the total MIMIC-CXR dataset (after filtering samples with "No Finding=1"). This subset served as our independent test set for all experiments.
2. **Direct Zero-Shot Inference:** For each sample in this 10,000-sample test set:
 - The raw radiology report text was extracted.
 - It was then processed through one of the defined preprocessing pipelines (e.g., Abbreviation Expansion + Noise Removal + Punctuation Removal + Lowercase Conversion, as detailed in Section 3.1).
 - The preprocessed text, along with an engineered prompt (e.g., Persona-based, Chain-of-Thought), was directly submitted to the pre-trained Large Language Model (GPT-4o-mini) via API.
 - The LLM generated predicted labels in a structured format (e.g., "label1, label2") without any prior exposure or training on this specific dataset or its labels.
3. **Metric Calculation:** The LLM's predicted labels for the 10,000 samples were directly compared against the ground-truth labels. Standard evaluation metrics (Accuracy, Precision, Recall, F1-score, and AUC) were then computed based on this comparison.
4. **Imbalance Management:** To ensure a fair assessment given the imbalanced label distribution within the selected 10,000-sample subset (as shown in Table 5), we applied the Macro-Averaging technique. This method calculates each metric independently for every class and then averages these values, ensuring all classes are given equal weight in the final evaluation.

The results presented in our tables (e.g., Table 6) are thus derived from this direct Zero-Shot inference on an independent, unseen test set.

5. Results

5.1. Evaluation of preprocessing, prompting, and section selection strategies

In this study, a total of 27 distinct experimental strategies were evaluated to systematically assess the impact of text preprocessing pipelines, prompt engineering techniques, uncertainty management, and report section selection on classification performance. The comprehensive results, evaluated on a random subset of 10,000 samples, are presented in Table 6.

The baseline configuration (Strategy 1) yielded an AUC of 0.80. As illustrated in the Receiver Operating Characteristic (ROC) curves in Fig 3, the model demonstrates high discriminative capability for distinct pathologies. For instance, the curves for "Consolidation" (AUC=0.97) and "Atelectasis" (AUC=0.92) bow sharply toward the top-left corner, indicating high sensitivity and specificity. Conversely, the curve for "Support Devices" (AUC=0.54) hovers near the diagonal random-guess line, highlighting a specific weakness in extracting this concept from the full, noisy report text using a basic prompt. This disparity underscores the necessity for the prompt optimization strategies explored in subsequent steps.

Table 6
Comparison of results from multiple strategies on a dataset with 10,000 random samples. The "GR type" column represents the method for managing uncertainty in –1 values in the Ground Truth. The best results are shown in Bold.

Strategy index	Text type	Preprocessing Index	Prompt index	accuracy	precision	recall	F1-score	AUC	GR type
1	complete	1	1	0.88	0.62	0.69	0.63	0.80	-1-1
2	complete	1	2	0.87	0.55	0.74	0.60	0.81	-1-1
3	complete	1	3	0.86	0.47	0.61	0.52	0.76	-1-1
4	complete	1	4	0.87	0.75	0.36	0.44	0.67	-1-1
5	complete	1	5	0.88	0.60	0.68	0.62	0.80	-1-1
6	complete	1	6	0.87	0.53	0.69	0.58	0.80	-1-1
7	complete	1	7	0.89	0.69	0.63	0.62	0.79	-1-1
8	complete	1	8	0.87	0.54	0.60	0.55	0.76	-1-1
9	complete	1	9	0.85	0.55	0.31	0.35	0.64	-1-1
10	complete	1	10	0.90	0.66	0.70	0.65	0.82	-1-1
11	Findings and Impression	1	1	0.89	0.57	0.79	0.64	0.85	-1-1
12	Findings	1	1	0.86	0.53	0.66	0.55	0.78	-1-1
13	Impression	1	1	0.96	0.76	0.85	0.80	0.92	-1-1
14	Impression	2	1	0.96	0.83	0.85	0.80	0.92	-1-1
15	Impression	3	1	0.93	0.64	0.76	0.64	0.86	-1-1
16	Impression	5	1	0.96	0.75	0.85	0.79	0.92	-1-1
17	Impression	6	1	0.96	0.84	0.84	0.80	0.91	-1-1
18	Impression	7	1	0.93	0.67	0.72	0.68	0.84	-1-1
19	Findings and Impression	7	1	0.95	0.74	0.84	0.77	0.90	-1-1
20	Findings	7	1	0.94	0.77	0.82	0.79	0.89	-1-1
21	Impression	7	1	0.96	0.88	0.84	0.85	0.92	-1-1
22	Findings and Impression	7	11	0.90	0.57	0.86	0.65	0.89	-1-1
23	Findings	7	11	0.86	0.51	0.61	0.52	0.75	-1-1
24	Impression	7	11	0.96	0.85	0.90	0.84	0.94	-1-1
25	Findings and Impression	7	1	0.92	0.58	0.86	0.67	0.89	-1-0
26	Findings	7	1	0.93	0.64	0.85	0.71	0.89	-1-0
27	Impression	7	1	0.94	0.72	0.85	0.76	0.90	-1-0

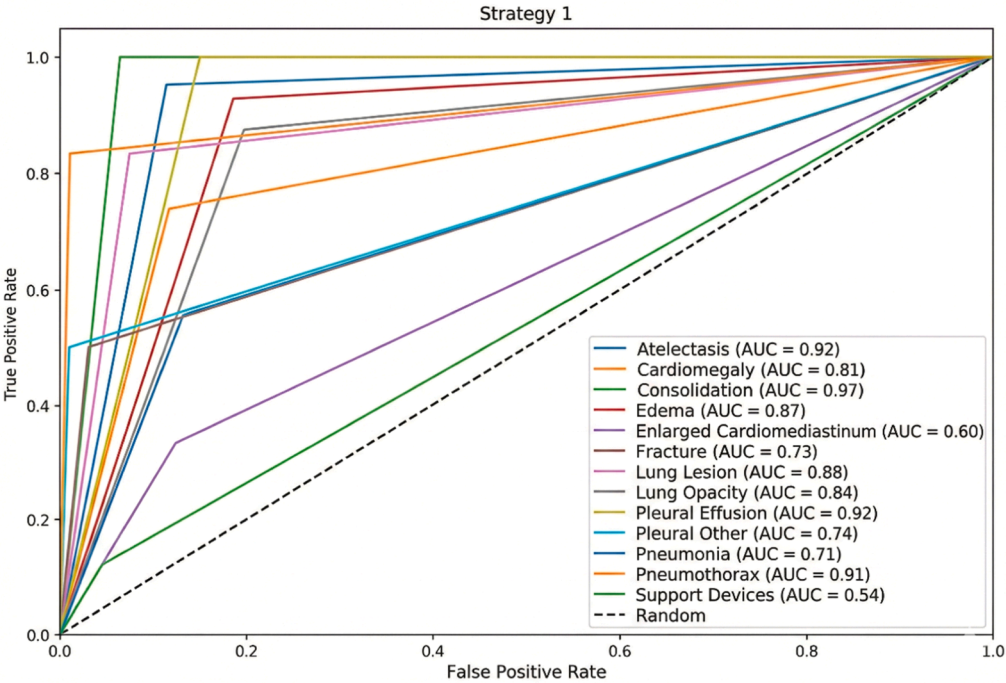


Fig. 3. ROC curve and AUC values in Strategy 1.

Further analysis (Strategies 11–21) demonstrated that constraining the input text to specific sections significantly impacted performance. Models utilizing the "IMPRESSION" section consistently outperformed those using "FINDINGS" or the complete text. The optimal performance was achieved in Strategy 21, which combined noise removal, abbreviation expansion, persona-based prompting, and mapping uncertain labels to positive (U-Ones). This configuration yielded the highest overall metrics: Accuracy = 0.96, Precision = 0.88, Recall = 0.84, F1-score

= 0.85, and AUC = 0.92.

The impact of section selection is visually summarized in Fig 4. Comparing the ROC curves, we observe a clear progression in model performance:

- Fig 4(a) (Combined Findings & Impression): Shows a robust performance with an average AUC of 0.90.

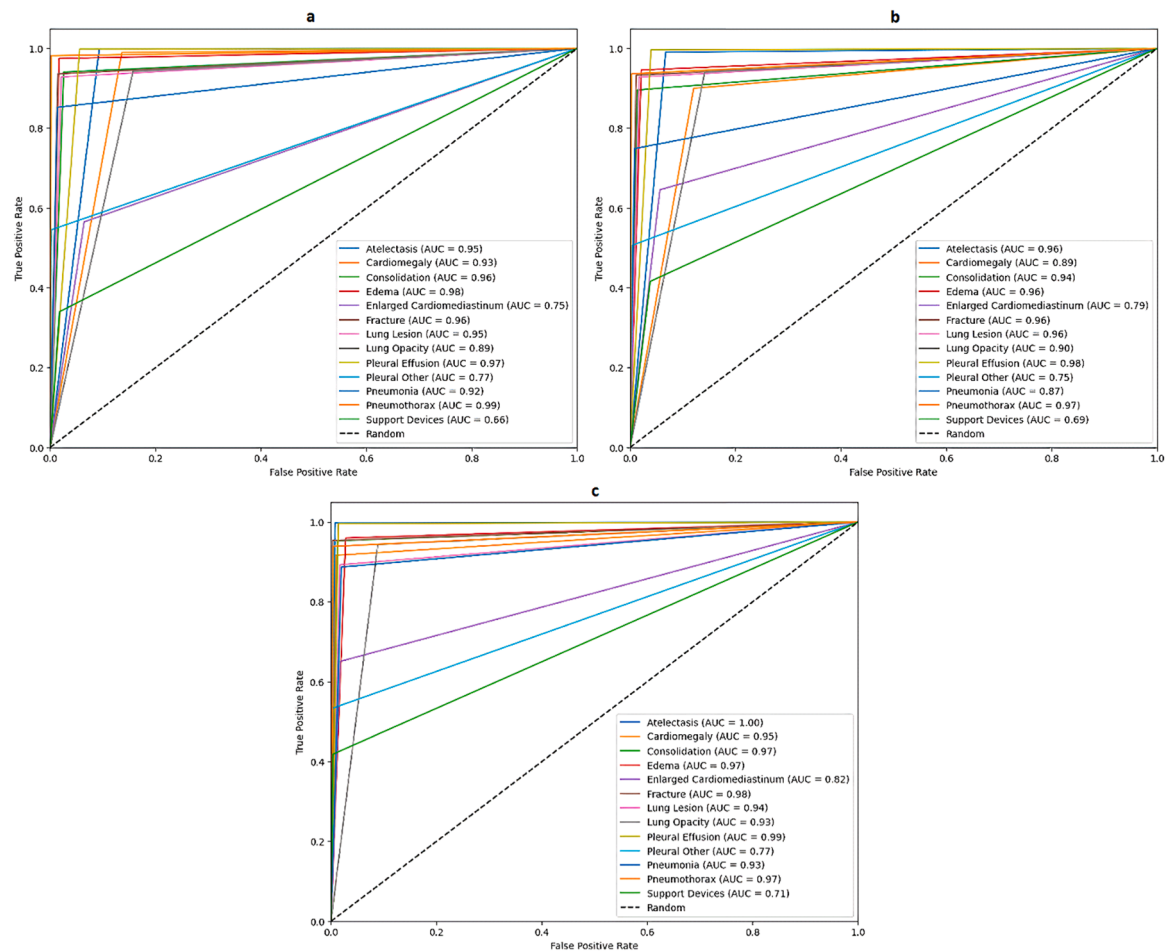


Fig. 4. ROC curve and AUC values for: a (text type "FINDINGS & IMPRESSION"), b (text type "FINDINGS"), and c (text type "IMPRESSION").

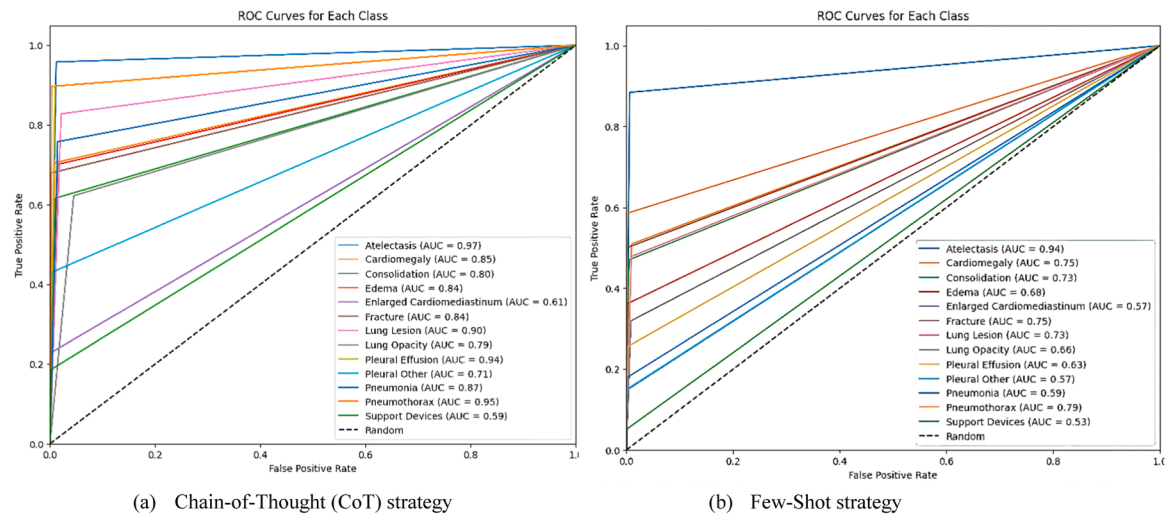


Fig. 5. Ablation study on prompt engineering strategies showing ROC curves and AUC values for: (a) Chain-of-Thought (CoT) strategy, achieving a mean AUC of 0.82; and (b) Few-Shot strategy, achieving a mean AUC of 0.69. Compared to the optimal Persona-based strategy (AUC 0.92, Fig. 4c), both CoT and Few-Shot approaches show degraded performance, with Few-Shot exhibiting significant loss in sensitivity across multiple pathologies.

- Fig 4(b) (Findings Only): Displays slightly lower AUCs (average 0.89), indicating that the verbose nature of the Findings section may introduce noise.
 - Fig 4(c) (Impression Only): Demonstrates the most optimal performance (Strategy 21). Notably, the curves for "Atelectasis" (AUC=1.00) and "Pleural Effusion" (AUC=0.99) indicate near-perfect classification.
- This visual comparison confirms that the "IMPRESSION" section, being a high-level summary, provides the most signal-rich data for the

LLM. Finally, altering the uncertainty management strategy by mapping uncertain labels to negative (Strategies 25–27) resulted in a noticeable decline in F1-scores compared to the primary approach.

5.2. Ablation study on prompt components

To isolate and quantify the individual contribution of specific prompt engineering techniques, we conducted a controlled ablation study based on the optimal configuration identified in Section 5.1 (Strategy 21). In this experiment, we held the preprocessing pipeline (Strategy 7), text selection ("IMPRESSION"), and uncertainty management (U-Ones) constant, while varying only the prompting strategy. We compared the Persona-based approach against Chain-of-Thought (CoT) and Few-Shot prompting.

The results reveal a significant performance hierarchy. The Persona-based prompting strategy (Strategy 21) yielded the highest performance, with an F1-score of 0.85 and an AUC of 0.92. This suggests that instructing the model to adopt the role of a radiologist effectively activates the relevant domain-specific latent knowledge required for accurate classification.

In contrast, the Chain-of-Thought (CoT) strategy achieved a lower F1-score of 0.72 and an AUC of 0.82. The ROC curves for the CoT strategy are presented in Fig. 5(a). While CoT is often effective for complex reasoning tasks, its lower performance here implies that for the "IMPRESSION" section which is already a synthesized summary of clinical reasoning, forcing the model to re-generate reasoning steps may introduce unnecessary noise or hallucination, degrading the final classification accuracy.

Most notably, the Few-Shot strategy resulted in a substantial performance drop, yielding an F1-score of 0.50 and an AUC of 0.69. The ROC curves for the Few-Shot strategy are shown in Fig. 5(b). The significantly low Recall (0.37) indicates that providing a limited set of examples (N-Shot) may have over-constrained the model. Instead of leveraging its broad general knowledge, the model likely overfit to the specific examples provided in the prompt, causing it to miss diverse presentations of pathologies that did not strictly match the few-shot examples. This finding underscores that for high-variability clinical tasks, a zero-shot Persona-based approach allows the LLM to generalize more effectively than a constrained Few-Shot approach.

5.3. Comparative benchmarking

To contextualize the performance of the proposed Zero-Shot framework, we conducted a two-phase comparative analysis against established benchmarks on the MIMIC-CXR dataset.

First, we compared our results against a diverse set of state-of-the-art models, including both image-based zero-shot methods and supervised text-based fine-tuning approaches. The comprehensive comparison is presented in Table 7.

The results indicate that the proposed method, which operates in a

zero-shot setting without prior training on MIMIC-CXR, achieves performance metrics comparable to or exceeding fully fine-tuned models. Specifically, our optimal configuration (Strategy 21) achieved an Accuracy of 0.96 and an AUC of 0.92. In contrast, the fine-tuning approach in [28] reported an accuracy of 0.87, and the zero-shot image-based approach in [32] reported an AUC of 0.73. This demonstrates that the extraction of clinical concepts from the "IMPRESSION" section using an LLM yields highly discriminative features for classification.

In the second phase, we performed a granular, label-by-label comparison against the most recent text classification baseline established by Uslu et al. [30], which utilizes BERT-based embeddings. Table 8 and Fig. 6 detail the F1-score distribution across the 13 target pathologies.

The proposed generative approach achieved a higher Mean F1-score of 0.85, compared to 0.80 for the BERT-based method [30]. Fig. 6 presents a radar chart comparing the F1-scores of our proposed strategies against the baseline [30]. The Green Line (Impression Strategy) consistently encompasses the Yellow Line (Baseline), visually indicating superior performance across the majority of classes. Notably, for challenging classes like "Consolidation" and "Pneumonia," our generative approach maintains high F1-scores (>0.90), whereas the discriminative baseline drops significantly.

However, it is noted that for the class "Support Devices," the proposed method recorded a lower F1-score (0.58) compared to the baseline (0.83). Despite this specific deviation, the overall variance of the F1-scores across all labels for the proposed method was 2.10, matching the variance of the baseline. This suggests that the model maintains consistent stability across diverse diagnostic categories while delivering higher average accuracy.

6. Discussion

6.1. Interpretability of prompt engineering and section selection

The experimental results underscore the critical importance of input text selection and prompt structure in clinical NLP pipelines. Our ablation study (Table 6) confirms that LLM performance is heavily dependent on the prompt strategy, acting as a decisive hyperparameter in the zero-shot framework. The superior performance achieved using the "IMPRESSION" section (AUC 0.92) compared to "FINDINGS" suggests that Large Language Models benefit significantly from the pre-synthesized diagnostic reasoning provided by radiologists. While the "FINDINGS" section contains granular observations, it often includes descriptive noise that may distract the model. In contrast, the "IMPRESSION" section represents a high-level summary where clinical reasoning has already been applied, aligning well with the LLM's capability to process abstract concepts.

Furthermore, the efficacy of Chain-of-Thought (CoT) prompting combined with Persona assignment (Strategy 21) highlights the model's reliance on structured reasoning. Simple task-specific prompts (Strategy 4) yielded significantly lower results (F1 = 0.44), whereas the

Table 7

Comparison of the proposed method's results with several previous studies on the MIMIC-CXR Database [19]. The best results are shown in Bold.

Method	Learning Approach	Text Type	# of labels	Mean Accuracy	Mean Precision	Mean Recall	Mean F1-score	Mean AUC
[26]	Cross-dataset zero-shot	—	5	0.50	—	—	—	—
[23]	Fine-tuning	FINDINGS & IMPRESSION	13	—	—	—	—	0.77
[28]	Fine-tuning	—	14	0.87	—	—	—	0.85
[30]	Fine-tuning	FINDINGS	14	—	—	—	0.80	—
[37]	Zero-shot	IMPRESSION	13	—	—	—	—	0.77
[32]	Zero-shot	IMPRESSION	13	—	—	—	—	0.73
Proposed Method	Zero-shot	FINDINGS & IMPRESSION	13	0.95	0.74	0.84	0.77	0.90
		FINDINGS		0.94	0.77	0.82	0.79	0.89
		IMPRESSION		0.96	0.88	0.84	0.85	0.92

Table 8
Comparison of F1-Score values for each label across different report text types of the proposed method and the study [30]. The best results are shown in Bold.

Labels	Method [30]	Proposed Method		
		FINDINGS & IMPRESSION	FINDINGS	IMPRESSION
Atelectasis	0.74	0.92	0.94	0.99
Cardiomegaly	0.76	0.76	0.81	0.94
Consolidation	0.56	0.82	0.84	0.91
Edema	0.85	0.96	0.93	0.94
Enlarged Cardiomediatinum	0.52	0.22	0.38	0.62
Fracture	0.61	0.79	0.83	0.93
Lung Lesion	0.67	0.81	0.81	0.80
Lung Opacity	0.79	0.85	0.86	0.88
Pleural Effusion	0.85	0.94	0.96	0.98
Pleural Other	0.47	0.60	0.53	0.63
Pneumonia	0.55	0.90	0.83	0.91
Pneumothorax	0.9	0.97	0.94	0.94
Support Devices	0.83	0.43	0.53	0.58
Mean	0.70	0.77	0.78	0.85
Variance	2.10	5.02	3.40	2.10

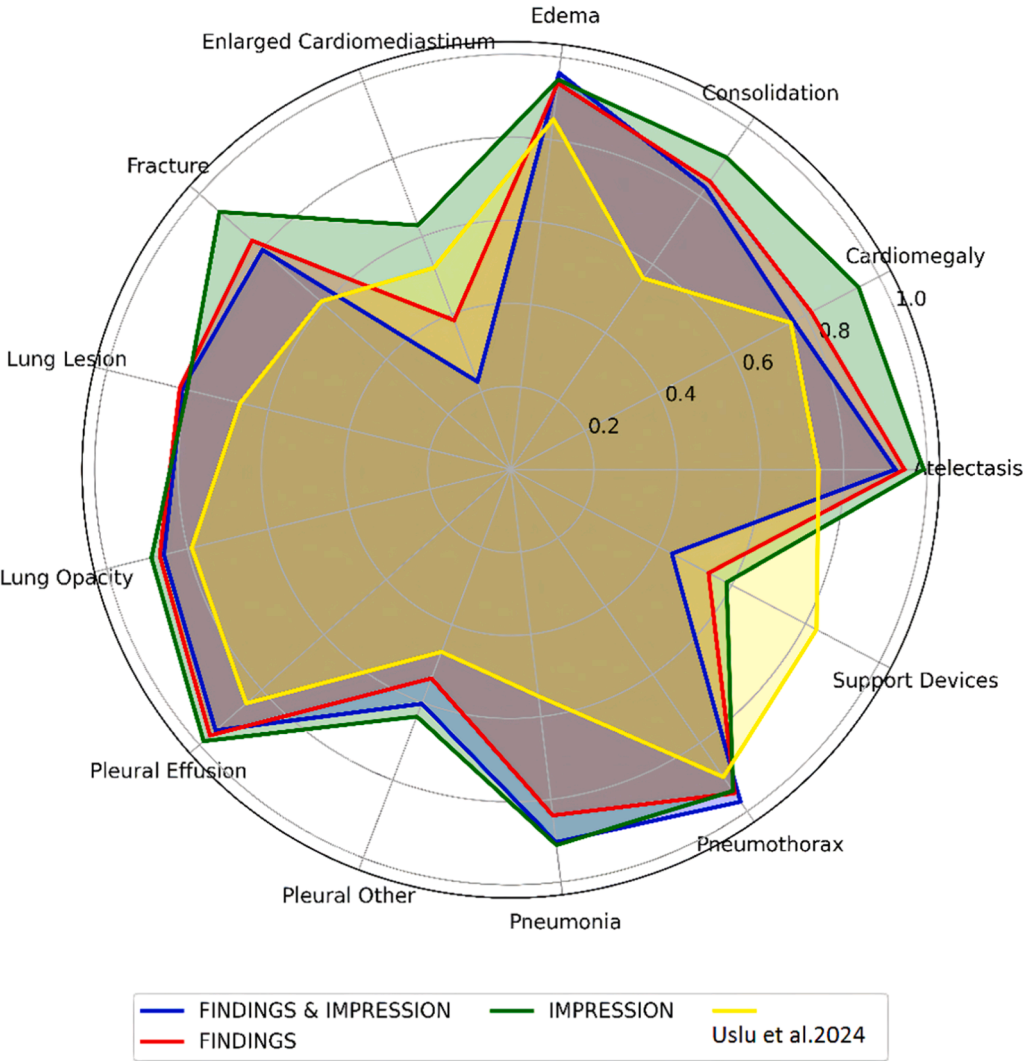


Fig. 6. Comparison of F1-Score values based on each label in different text report scenarios of the proposed method and the study [30].

engineered ensemble strategy achieved SOTA performance (F1 =0.85). This variance highlights that while LLMs possess the latent knowledge required for diagnosis, "Prompt Engineering" is the essential mechanism required to unlock this potential. By systematically tuning these

prompts, we effectively activate domain-specific latent knowledge, mirroring the cognitive process of human experts who synthesize disparate observations into a coherent diagnosis.

6.2. Comparative analysis with encoder-based and image-based models

The comparison with state-of-the-art text classifiers reveals a fundamental shift in the capabilities of NLP architectures. Previous benchmarks established by Uslu et al. [30] utilizing BERT-based encoders (e.g., ClinicalBERT, CXR-BERT) achieved an F1-score of 0.80. Our generative approach with GPT-4o-mini achieved 0.85. This performance gap can be attributed to the architectural differences: BERT-based models are Discriminative (encoding text into vectors for classification heads), whereas GPT-based models are Generative (capable of following complex instructions and leveraging world knowledge). The lower variance in our results (2.10) compared to the baseline's variable performance further suggests that generative models are less brittle when handling rare or complex pathologies.

It is also pertinent to position this work relative to image-based and multi-modal models. As shown in Table 7, our method's AUC of 0.90 (Findings & Impression) and 0.92 (Impression) outperforms the zero-shot image-based baselines, such as the cross-dataset zero-shot approach in [26] (Accuracy 0.50) and other zero-shot benchmarks on IMPRESSION text reported as 0.77 [37] and 0.73 [32]. While supervised image classification models using advanced architectures like MaxViT [15] and Swin Transformers [16] may achieve higher accuracies in fully supervised settings, our text-based Zero-Shot approach offers a distinct advantage: Label Efficiency. Unlike supervised models that require thousands of annotated images, our framework functions without any training data, offering a scalable solution for generating preliminary labels or structuring retrospective clinical data where images may be unavailable or computationally expensive to process.

6.3. Disentangling model capability from prompt efficacy

A critical question in deploying Large Language Models for clinical applications is distinguishing between the inherent knowledge of the pre-trained foundation model and the efficacy of the prompting strategy employed. Our experimental design allows us to rigorously isolate these factors. Since the underlying model (GPT-4o-mini) was held constant across all 27 experimental strategies, any variance in performance metrics is directly attributable to the methodology of interaction—specifically, the prompt engineering and text selection strategies.

As demonstrated in the ablation results (Table 6), the baseline performance using the model with a naive, task-specific prompt (Strategy 9) yielded an F1-score of only 0.35. In stark contrast, by applying our proposed framework which integrates Persona-based instructions, Chain-of-Thought reasoning, and the targeted selection of the "IMPRESSION" section (Strategy 21), the performance surged to 0.85. This 50-percentage point absolute improvement serves as robust empirical evidence that the model's raw capability, while necessary, is insufficient for high-precision clinical tasks without structured guidance.

This finding has significant implications for the deployment of generative AI in healthcare. It suggests that "Prompt Engineering" acts as a decisive hyperparameter, analogous to weight fine-tuning in traditional deep learning but without the computational cost. The ability to unlock SOTA performance from a general-purpose model through purely textual instructions validates the potential for rapid, low-resource adaptation of LLMs to specialized medical domains, provided the prompting framework is rigorously designed and validated.

6.4. Uncertainty management and limitations

A critical challenge in classifying medical reports is the ambiguity inherent in clinical language, often annotated as "-1" in datasets like CheXpert and MIMIC-CXR. Our experiments demonstrated that the "U-Ones" strategy (mapping uncertainty to positive class 1) significantly outperformed the "U-Zeros" strategy (mapping to 0) in terms of recall.

From a computational perspective, this suggests that the features of "uncertain" cases in the text space share more semantic similarity with "positive" cases than "negative" ones.

Clinically, this aligns with the "Recall-First" principle in medical screening, where the cost of a False Negative (missing a disease) is substantially higher than a False Positive (further testing required). However, we acknowledge that binarizing uncertainty is a simplification. The "-1" label often represents a spectrum of clinical confidence, ranging from "cannot exclude" to "likely." While supervised learning approaches can employ techniques like Label Smoothing or Loss Relaxation [23] to model this soft probability, zero-shot frameworks face a unique constraint: the output must be discrete text generated by the LLM.

Therefore, while our "U-Ones" strategy is optimal for algorithmic benchmarking, its direct impact on clinical decision-making requires further validation. Future systems should aim to preserve this uncertainty rather than resolving it. To this end, we propose moving beyond binary mapping by integrating Evidence Theory (Dempster-Shafer Theory) [63] into the prompt engineering pipeline. By instructing the LLM to output a "Confidence Score" or "Belief Mass" alongside the label, future systems could quantify the degree of uncertainty directly from the report text, allowing for a more nuanced decision-support system where "Uncertain" remains a valid, actionable output for radiologist review.

6.5. Generalizability and methodological constraints

While this study utilizes the MIMIC-CXR dataset as a primary benchmark due to its scale and public availability, the proposed framework is designed to be inherently generalizable. Unlike supervised deep learning models that often suffer from overfitting to specific hospital reporting styles or demographic distributions, the proposed Zero-Shot approach relies on the semantic reasoning capabilities of Large Language Models. Because the model leverages general medical knowledge rather than learning statistical patterns from a specific training set, it is expected to maintain robustness when applied to reports from different clinical institutions. However, we acknowledge that the performance is intrinsically linked to the capabilities of the selected LLM. In this study, we utilized a single model, GPT-4o-mini, to establish a baseline for efficiency and accuracy. Future research should extend this framework to include a broader range of Large Language Models, including open-source alternatives, and evaluate performance across diverse datasets to empirically validate cross-domain generalizability.

7. Conclusion

Accurate classification of chest radiology reports is a fundamental challenge in clinical Natural Language Processing, significantly hindered by the scarcity of large scale labeled datasets and the prohibitive costs of expert annotation. This study evaluated the efficacy of pre trained Large Language Models within a Prompt based Zero Shot Learning Framework for the multi label classification of 13 pathologies from the MIMIC CXR dataset. By focusing exclusively on textual reports, our approach eliminates the need for direct image processing and the associated computational overhead of training visual models.

We systematically investigated the impact of various text pre-processing pipelines and prompt engineering strategies. Our findings demonstrate that rigorous text normalization combined with structured prompting techniques, specifically Persona based instructions and Chain of Thought reasoning, can significantly enhance LLM comprehension and inference accuracy. A key finding of this research is the identification of the optimal information source: analyzing the full report is unnecessary. The IMPRESSION section alone, which synthesizes clinical reasoning, contains sufficient signal to achieve state of the art results. Our optimal configuration achieved a weighted average F1 score of 0.85 and an AUC of 0.92, significantly outperforming both unsupervised baselines and traditional BERT based supervised classifiers.

The broader significance of this work lies in demonstrating that Generative LLMs can effectively bypass the annotation bottleneck. We provide empirical evidence that zero shot frameworks can achieve performance competitive with, or superior to, fully supervised models without requiring a single labeled training example. This paradigm shift offers a scalable pathway for developing cost effective Clinical Decision Support Systems that can be rapidly deployed across institutions without the need for site specific model retraining.

Despite these promising results, certain limitations remain. Our evaluation was conducted on a random subset of 10,000 samples due to API constraints, and we focused on a specific set of 13 labels. Additionally, reliance on text alone assumes the accuracy of the original radiologist report; visual features not documented in the text are naturally lost. Furthermore, the binary handling of uncertain labels simplifies the clinical reality. To address these challenges, future work should focus on:

1. **Uncertainty Quantification:** Integrating data fusion methods, such as Dempster Shafer Theory, to provide a probabilistic measure of confidence rather than a binary output.
2. **Model Diversity:** Evaluating the framework on open source LLMs, such as Llama 3 or Mistral, to assess cost efficiency and privacy compliance.
3. **Multimodal Integration:** Using this text-based engine as a Teacher model to guide the training of lightweight image encoders, creating a holistic multimodal diagnostic system.

CRedit authorship contribution statement

Hassan Katal: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Jamal Ghaseemi:** Supervision, Writing – review & editing, Methodology, Investigation, Conceptualization. **Meysam Roostaee:** Supervision, Writing – review & editing, Methodology, Investigation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

The manuscript text has been reviewed using AI-powered proof-reading tools. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets supporting the conclusions of this article are available in the physionet repository, <https://physionet.org/content/mimic-cxr/2.1.0/> and <https://physionet.org/content/mimic-cxr-jpg/2.1.0/>.

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